

OPS RUNBOOK V1

Section 1 – Operational Governance Framework

1.1 Purpose

Establish the governance structure governing day-to-day operations of the Transparent Opportunity Decision Platform (TODP) production environment.

1.2 Objectives

- Ensure operational consistency across all environments.
- Define operational accountability structures.
- Establish operational oversight mechanisms.
- Support enterprise reliability and compliance requirements.
- Enable auditability and operational traceability.

1.3 Scope

Applies to:

- Production operations
- Platform administration
- Service management
- Incident management
- Change management
- Release operations
- Monitoring operations
- Security operations
- Business continuity activities

Covered Components:

- Opportunity Module
- Authority Module
- Evidence Module
- Retrieval Module
- Ranking Module
- Recommendation Module
- Governance Module
- Audit Module
- Replay Module

1.4 Governance Principles

- Transparency
- Explainability
- Traceability
- Accountability
- Reliability
- Security
- Operational Excellence
- Continuous Improvement

1.5 Operational Governance Structure

Executive Oversight

→ Product Governance Board

Operational Oversight

→ Platform Operations Committee

Execution Layer

- Platform Operations Team
- Security Operations Team
- DevOps Team
- Data Operations Team

Assurance Layer

- Internal Audit
- Compliance Team
- Risk Management Function

1.6 Governance Responsibilities

Product Governance Board

- Strategic oversight
- Risk review
- Operational policy approval

Platform Operations Committee

- Monthly operational review
- SLA review
- Incident trend review
- Capacity review

Operations Team

- Service execution

- Monitoring
- Incident response

Security Team

- Threat monitoring
- Security investigations
- Vulnerability response

1.7 Operational Policies

Operations shall comply with:

- Production Access Policy
- Change Management Policy
- Incident Response Policy
- Backup Policy
- Disaster Recovery Policy
- Security Operations Policy
- Data Governance Policy

1.8 Operational Review Cadence

Daily

- System health review
- Incident review

Weekly

- Capacity review
- Change review

Monthly

- SLA review
- Risk review
- Security review

Quarterly

- Governance assessment
- Operational maturity assessment

1.9 Metrics

- Platform Availability
- MTTR
- MTTD
- Incident Volume
- Change Success Rate
- Security Event Volume
- Capacity Utilization

1.10 Audit Requirements

All operational activities shall generate:

- Audit records
- Operational logs
- Escalation records
- Change records
- Incident records

1.11 Traceability

RTM References:

- OPS-GOV-001
- OPS-GOV-002
- GOV-OPS-001
- AUD-OPS-001

1.12 Ownership

Primary Owner:

Head of Platform Operations

Review Authority:

Platform Operations Committee

Approval Authority:

Product Governance Board

1.13 Section Summary

This section defines the operational governance structure, oversight model, responsibilities, review processes, and governance controls governing TODP operations.

OPS RUNBOOK V1

Section 2 – Production Operations Model

2.1 Purpose

Define the operational execution model for maintaining production services across TODP environments.

2.2 Objectives

- Ensure stable production operations.
- Maintain service reliability.
- Support enterprise SLA commitments.
- Enable controlled operational execution.

2.3 Production Service Model

Users

→ API Gateway

→ Application Services

→ Retrieval Services

→ Ranking Services

→ Recommendation Services

→ Governance Services

→ Audit Services

→ Data Stores

2.4 Operational Domains

Domain A

Application Operations

Domain B

Infrastructure Operations

Domain C

Security Operations

Domain D

Data Operations

Domain E

Platform Reliability Operations

2.5 Service Ownership

Opportunity Services

Owner: Application Operations

Authority Services

Owner: Application Operations

Evidence Services

Owner: Application Operations

Retrieval Services

Owner: Platform Reliability Team

Ranking Services

Owner: Platform Reliability Team

Recommendation Services

Owner: Application Operations

Governance Services

Owner: Governance Operations

Audit Services

Owner: Governance Operations

Replay Services

Owner: Platform Reliability Team

2.6 Operating Model Principles

- Automation First
- Monitoring by Default
- Security by Default
- Controlled Change
- Evidence-Based Operations
- Auditable Actions

2.7 Operational Activities

Continuous Activities

- Monitoring
- Alert Review
- Capacity Observation
- Log Analysis

Scheduled Activities

- Patch Management
- Backup Validation
- Capacity Planning
- Security Review

2.8 Operational Shift Model

Business Hours Coverage

Operations Team

Extended Coverage

On-call Engineers

Critical Coverage

24x7 Incident Response Rotation

2.9 Operational Workflows

Monitoring

→ Detection

→ Classification

→ Escalation

→ Resolution

→ Closure

2.10 SLA Alignment

Availability Target:

99.9%

Incident Acknowledgement:

Defined by Severity

Resolution Targets:

Governed by Incident Management Process

2.11 Compliance Requirements

Operations must support:

- SOC2 Controls
- ISO27001 Controls
- Internal Governance Standards
- Audit Requirements

2.12 Reporting Requirements

Daily Report

- Health Status

Weekly Report

- Incident Summary

Monthly Report

- SLA Performance

Quarterly Report

- Operational Risk Review

2.13 Operational Constraints

- No direct production modifications without approval.
- All production changes require traceable records.
- Emergency actions require post-event review.

2.14 Traceability

RTM References:

- OPS-MOD-001
- OPS-MOD-002
- SLA-001
- AUD-OPS-002

2.15 Section Summary

This section defines the operational execution framework supporting reliable production operation of TODP services.

OPS RUNBOOK V1

Section 3 – Service Ownership & Responsibility Matrix

3.1 Purpose

Define ownership, accountability, and operational responsibility across TODP services.

3.2 Objectives

- Eliminate ownership ambiguity.
- Support rapid incident resolution.
- Enable governance accountability.
- Establish operational accountability.

3.3 Responsibility Framework

Model:

RACI

R = Responsible

A = Accountable

C = Consulted

I = Informed

3.4 Ownership Categories

- Product Ownership
- Platform Ownership
- Service Ownership
- Security Ownership
- Data Ownership
- Compliance Ownership

3.5 Responsibility Matrix

Opportunity Services

R: Application Team

A: Product Owner

C: Governance Team

I: Operations Team

Authority Services

R: Application Team

A: Product Owner

C: Governance Team

I: Operations Team

Evidence Services

R: Application Team

A: Product Owner

C: Data Team

I: Operations Team

Retrieval Services

R: Platform Team

A: Platform Lead

C: Data Team

I: Governance Team

Ranking Services

R: Platform Team

A: Platform Lead

C: Product Team

I: Governance Team

Recommendation Services

R: Application Team

A: Product Owner

C: Governance Team

I: Operations Team

Governance Services

R: Governance Team

A: Governance Lead

C: Compliance Team

I: Executive Stakeholders

Audit Services

R: Governance Team

A: Governance Lead

C: Compliance Team

I: Executive Stakeholders

Replay Services

R: Platform Team

A: Platform Lead

C: Governance Team

I: Operations Team

3.6 Infrastructure Ownership

Cloud Infrastructure

Owner: DevOps Team

Network Services

Owner: DevOps Team

Monitoring Platform

Owner: Operations Team

Logging Platform

Owner: Operations Team

3.7 Security Ownership

IAM

Security Team

Threat Monitoring

Security Team

Vulnerability Management

Security Team

3.8 Escalation Ownership

L1

Operations Team

L2

Platform Engineering

L3

Product Engineering

L4

Executive Governance Review

3.9 Accountability Requirements

- All systems require named owners.
- Ownership shall be documented.
- Ownership shall be reviewed quarterly.

3.10 Audit Requirements

Auditors must be able to identify:

- Service Owner
- Backup Owner
- Escalation Owner
- Governance Owner

3.11 Metrics

- Ownership Coverage
- Escalation Compliance

- Resolution Accountability

3.12 Traceability

RTM References:

- OWN-001
- OWN-002
- GOV-RESP-001

3.13 Section Summary

This section defines ownership and accountability structures supporting operational governance and service management.

OPS RUNBOOK V1

Section 4 – Operational Escalation Framework

4.1 Purpose

Establish the escalation process for operational, security, compliance, and platform events.

4.2 Objectives

- Accelerate issue resolution.
- Ensure appropriate stakeholder involvement.
- Reduce operational risk.
- Maintain governance compliance.

4.3 Escalation Principles

- Timeliness
- Accountability
- Traceability
- Severity-Based Response
- Evidence Preservation

4.4 Escalation Categories

Category A

Operational Incident

Category B

Security Incident

Category C

Data Incident

Category D

Compliance Incident

Category E

Infrastructure Incident

4.5 Escalation Levels

Level 1

Operations Team

Level 2

Service Owner

Level 3

Platform Leadership

Level 4

Executive Governance Board

4.6 Severity Classification

SEV-1

Critical Service Outage

SEV-2

Major Service Degradation

SEV-3

Moderate Impact

SEV-4

Minor Impact

SEV-5

Informational Event

4.7 Escalation Triggers

Availability Breach

Security Detection

Unauthorized Access Event

Data Integrity Concern

Compliance Violation

Capacity Exhaustion

Backup Failure

Disaster Recovery Failure

4.8 Escalation Workflow

Detection

→ Classification

→ Severity Assignment

→ Notification

→ Escalation

→ Resolution

→ Post-Incident Review

4.9 Notification Requirements

SEV-1

Immediate

SEV-2

Within Defined Operational Window

SEV-3

Business Hours

SEV-4

Routine Processing

SEV-5

Operational Logging

4.10 Executive Escalation Criteria

Mandatory executive notification:

- Major outage
- Security breach
- Regulatory exposure
- Data loss event
- Significant customer impact

4.11 Escalation Documentation

Required records:

- Incident ID
- Escalation ID
- Timeline
- Decision Log
- Resolution Evidence

4.12 Audit Requirements

All escalations shall produce:

- Immutable audit records
- Notification evidence
- Resolution evidence
- Postmortem documentation

4.13 Metrics

- Escalation Time
- Resolution Time
- Escalation Accuracy
- SLA Compliance
- Repeat Incident Rate

4.14 Governance Requirements

Governance review required for:

- SEV-1 incidents
- Compliance incidents
- Security incidents
- Audit findings

4.15 Traceability

RTM References:

- ESC-001
- ESC-002
- INC-001
- AUD-ESC-001

4.16 Section Summary

This section defines the operational escalation framework, severity model, notification requirements, governance controls, and escalation accountability mechanisms supporting TODP production operations.

OPS RUNBOOK V1

Section 5 – Incident Management Framework

5.1 Purpose

Establish a standardized framework for identifying, managing, resolving, documenting, and reviewing incidents affecting TODP services.

5.2 Objectives

- Minimize service disruption.
- Restore service within SLA targets.
- Ensure consistent incident handling.
- Preserve operational evidence.
- Support governance and audit requirements.

5.3 Scope

Applies to:

- Application incidents
- Infrastructure incidents
- Security incidents
- Data incidents
- Third-party service incidents
- Compliance incidents

5.4 Incident Management Principles

- Rapid Detection
- Controlled Response
- Accurate Classification
- Traceable Decision Making
- Root Cause Identification
- Continuous Improvement

5.5 Incident Lifecycle

Detection

→ Validation

→ Classification

→ Prioritization

→ Assignment

→ Containment

→ Resolution

→ Verification

→ Closure

→ Postmortem

5.6 Severity Model

SEV-1 Critical

Complete service outage, data compromise, regulatory exposure.

SEV-2 High

Major functionality unavailable, significant customer impact.

SEV-3 Medium

Partial degradation with workaround available.

SEV-4 Low

Minor service impact.

SEV-5 Informational

No service impact.

5.7 Incident Roles

Incident Commander

- Overall coordination

Technical Lead

- Resolution ownership

Communications Lead

- Stakeholder communication

Governance Representative

- Compliance oversight

Scribe

- Timeline documentation

5.8 Response Requirements

SEV-1

Immediate engagement

SEV-2

Priority response

SEV-3

Business-hour response

SEV-4

Scheduled response

SEV-5

Operational tracking

5.9 Communication Requirements

Internal Stakeholders

- Operations Team
- Engineering Teams
- Governance Team

External Stakeholders

- Customers
- Partners
- Regulators (if required)

5.10 Postmortem Requirements

Mandatory for:

- All SEV-1 incidents
- All SEV-2 incidents
- Security incidents
- Compliance incidents

5.11 Root Cause Analysis Requirements

RCA must include:

- Event chronology
- Contributing factors
- Root cause determination
- Corrective actions
- Preventive actions

5.12 Audit Requirements

Incident records must contain:

- Incident ID
- Severity
- Timeline
- Escalations
- Resolution evidence
- RCA documentation

5.13 Metrics

- MTTR
- MTTD
- Incident Volume
- Repeat Incidents
- SLA Compliance

5.14 Traceability

RTM References:

- INC-001
- INC-002
- GOV-INC-001
- AUD-INC-001

5.15 Ownership

Primary Owner:

Head of Platform Operations

Review Authority:

Platform Operations Committee

5.16 Section Summary

This section defines the incident management lifecycle, governance controls, responsibilities, escalation requirements, and audit obligations for TODP operations.

OPS RUNBOOK V1

Section 6 – Problem Management & Root Cause Governance

6.1 Purpose

Establish a formal process for identifying and eliminating underlying causes of recurring incidents.

6.2 Objectives

- Reduce incident recurrence.
- Improve platform stability.
- Strengthen operational resilience.
- Support continuous improvement.

6.3 Scope

Applies to:

- Recurring incidents
- High-severity incidents
- Systemic failures
- Operational risks

- Security weaknesses

6.4 Problem Management Principles

- Prevention over reaction
- Data-driven analysis
- Cross-functional collaboration
- Measurable remediation

6.5 Problem Lifecycle

Problem Identification

- Analysis
- Root Cause Determination
- Remediation Planning
- Implementation
- Validation
- Closure

6.6 Problem Sources

- Incident trends
- Monitoring insights
- Security findings
- Audit findings
- Capacity reviews
- Customer complaints

6.7 Problem Classification

Category A

Application

Category B

Infrastructure

Category C

Security

Category D

Data

Category E

Governance

6.8 Root Cause Analysis Methods

- Five Whys
- Fault Tree Analysis
- Timeline Analysis
- Dependency Mapping
- Impact Analysis

6.9 Remediation Requirements

Each problem record shall include:

- Root cause
- Impact assessment
- Risk assessment
- Corrective actions
- Target completion date
- Responsible owner

6.10 Governance Requirements

Governance review required for:

- Critical problems
- Security-related problems
- Regulatory-impacting problems

6.11 Validation Requirements

Before closure:

- Fix verification completed
- Monitoring validation completed
- Stakeholder approval obtained

6.12 Reporting Requirements

Monthly Reports:

- Open problems
- Closed problems

- Recurrence rates
- Remediation effectiveness

6.13 Metrics

- Problem Backlog
- Problem Resolution Time
- Repeat Incident Rate
- RCA Completion Rate

6.14 Audit Requirements

Maintain:

- Problem Register
- RCA Documentation
- Remediation Records
- Validation Evidence

6.15 Traceability

RTM References:

- PROB-001
- PROB-002
- AUD-PROB-001

6.16 Section Summary

This section establishes governance and operational controls for identifying, analyzing, correcting, and preventing systemic operational issues.

Section 7 – Change Management Operations

7.1 Purpose

Define the operational process governing modifications to production systems and services.

7.2 Objectives

- Minimize change-related failures.
- Ensure controlled deployments.
- Maintain service stability.
- Preserve operational traceability.

7.3 Scope

Applies to:

- Infrastructure changes
- Application changes
- Security changes
- Configuration changes
- Data schema changes
- Integration changes

7.4 Change Management Principles

- Approval before execution
- Risk-based assessment

- Rollback readiness
- Complete auditability

7.5 Change Categories

Standard Change

Pre-approved low-risk changes

Normal Change

Risk-assessed scheduled changes

Emergency Change

Immediate implementation required

7.6 Change Lifecycle

Request

→ Impact Assessment

→ Risk Assessment

→ Approval

→ Scheduling

→ Implementation

→ Validation

→ Closure

7.7 Change Advisory Board (CAB)

Responsibilities:

- Review changes
- Assess risks
- Approve implementation
- Monitor outcomes

7.8 Change Request Requirements

Required Fields:

- Change ID
- Description
- Business justification
- Risk level
- Rollback plan
- Validation plan

7.9 Risk Classification

Low Risk

Medium Risk

High Risk

Critical Risk

7.10 Emergency Change Governance

Requirements:

- Incident linkage
- Executive notification
- Post-implementation review
- Governance approval after execution

7.11 Validation Requirements

Post-change validation shall verify:

- Service availability
- Functional integrity
- Security posture
- Data integrity

7.12 Rollback Requirements

Every production change shall include:

- Rollback procedure
- Rollback owner
- Rollback decision criteria

7.13 Metrics

- Change Success Rate
- Emergency Change Volume
- Failed Changes

- Rollback Frequency

7.14 Audit Requirements

Maintain:

- Change Requests
- Approval Records
- Validation Records
- Rollback Records

7.15 Traceability

RTM References:

- CHG-001
- CHG-002
- OPS-CHG-001
- AUD-CHG-001

7.16 Section Summary

This section establishes the governance model, approval controls, risk management requirements, and operational execution standards for production changes.

OPS RUNBOOK V1

Section 8 – Release & Deployment Operations

8.1 Purpose

Define operational procedures governing software releases and production deployments.

8.2 Objectives

- Deliver reliable releases.
- Reduce deployment risk.
- Maintain deployment traceability.
- Support controlled service evolution.

8.3 Scope

Applies to:

- Application releases
- Infrastructure releases
- Configuration releases
- Security updates
- Emergency deployments

8.4 Release Principles

- Automation First
- Controlled Promotion
- Repeatable Execution
- Rollback Readiness
- Auditability

8.5 Environment Promotion Model

Development

→ Integration

→ QA

→ UAT

→ Staging

→ Production

8.6 Release Lifecycle

Build

→ Test

→ Security Validation

→ Approval

→ Deployment

→ Verification

→ Monitoring

→ Closure

8.7 Release Approval Requirements

Required Approvals:

- Product Owner
- Engineering Lead
- Operations Lead
- Security Representative (where applicable)

8.8 Deployment Methods

- Blue-Green Deployment

- Rolling Deployment
- Canary Deployment
- Emergency Deployment

8.9 Deployment Validation

Verify:

- Service health
- API functionality
- Data integrity
- Monitoring coverage
- Security controls

8.10 Rollback Governance

Rollback triggers include:

- SLA degradation
- Service outage
- Security concern
- Data inconsistency

8.11 Release Documentation

Required Records:

- Release ID
- Deployment ID
- Version Information

- Change References
- Validation Results

8.12 Post-Release Monitoring

Monitoring Period:

- Critical Releases: 24–72 hours
- Standard Releases: Defined operational window

8.13 Metrics

- Deployment Success Rate
- Release Frequency
- Failed Deployment Rate
- Rollback Rate
- Deployment Duration

8.14 Audit Requirements

Maintain:

- Release Records
- Deployment Logs
- Approval Evidence
- Validation Reports

8.15 Traceability

RTM References:

- REL-001
- REL-002
- DEP-001
- AUD-REL-001

8.16 Section Summary

This section defines release governance, deployment controls, approval requirements, validation procedures, rollback standards, and audit obligations supporting reliable TODP service delivery.

OPS RUNBOOK V1

Section 9 – Monitoring & Observability Operations

9.1 Purpose

Establish monitoring, observability, alerting, and operational visibility standards required to maintain reliable operation of the Transparent Opportunity Decision Platform (TODP).

9.2 Objectives

- Detect service degradation proactively.
- Improve operational visibility.
- Reduce Mean Time to Detect (MTTD).
- Support incident response activities.
- Enable audit-ready operational evidence.

9.3 Scope

Applies to:

- Application services
- APIs
- Infrastructure

- Databases
- Search systems
- Message queues
- Security services
- Governance services
- Audit services

9.4 Observability Principles

- Monitoring by Default
- Alerting by Exception
- End-to-End Visibility
- Measurable Reliability
- Traceable Events
- Operational Transparency

9.5 Observability Architecture

Application Layer

→ Metrics Collection

Infrastructure Layer

→ Metrics Collection

Security Layer

→ Event Collection

Governance Layer

→ Audit Collection

Central Observability Platform

→ Dashboards

→ Alerts

→ Reporting

9.6 Monitoring Domains

Domain A

Availability Monitoring

Domain B

Performance Monitoring

Domain C

Security Monitoring

Domain D

Infrastructure Monitoring

Domain E

Business Process Monitoring

Domain F

Data Quality Monitoring

9.7 Monitoring Requirements

All production services shall provide:

- Health endpoints
- Operational metrics
- Structured logs
- Trace identifiers
- Audit events

9.8 Alert Classification

Critical

High

Medium

Low

Informational

9.9 Alert Handling Workflow

Alert Generated

→ Validation

→ Classification

→ Assignment

→ Investigation

→ Resolution

→ Closure

9.10 Dashboard Requirements

Executive Dashboard

Operations Dashboard

Security Dashboard

Governance Dashboard

Platform Reliability Dashboard

9.11 Service Health Indicators

- Service Availability
- API Latency
- Error Rate
- Resource Utilization
- Queue Backlog
- Search Performance
- Recommendation Latency

9.12 Monitoring Retention

Metrics:

13 Months

Operational Logs:

Minimum 12 Months

Audit Logs:

Per Security Architecture Requirements

9.13 Metrics

- Alert Volume
- Alert Accuracy
- MTTD
- False Positive Rate
- Monitoring Coverage

9.14 Audit Requirements

Maintain:

- Alert Records
- Dashboard Evidence
- Monitoring Configurations
- Escalation Records

9.15 Traceability

RTM References:

- MON-001
- MON-002
- OBS-001
- AUD-MON-001

9.16 Ownership

Primary Owner:

Platform Operations Team

Review Authority:

Platform Operations Committee

9.17 Section Summary

This section defines monitoring, observability, alerting, dashboard governance, operational visibility requirements, and audit controls supporting reliable TODP operations.

OPS RUNBOOK V1

Section 10 – Logging & Operational Evidence Management

10.1 Purpose

Define requirements for operational logging, evidence preservation, and traceability across TODP production environments.

10.2 Objectives

- Maintain operational traceability.
- Support investigations.
- Enable compliance audits.
- Preserve decision evidence.
- Support replayability requirements.

10.3 Scope

Applies to:

- Application logs
- Infrastructure logs
- Security logs
- Governance logs
- Audit logs
- Deployment logs
- Access logs

10.4 Logging Principles

- Structured Logging
- Immutable Auditability
- Timestamp Accuracy
- Evidence Preservation

- Security Protection

10.5 Logging Categories

Category A

Application Logs

Category B

Infrastructure Logs

Category C

Security Logs

Category D

Governance Logs

Category E

Audit Logs

Category F

Operational Logs

10.6 Mandatory Log Attributes

Every log record shall include:

- Event ID

- Timestamp
- Correlation ID
- Service Name
- Environment
- Severity
- Actor Identifier
- Event Type

10.7 Correlation Requirements

The following identifiers shall be traceable:

Opportunity ID

Authority ID

Evidence ID

Retrieval ID

Ranking ID

Recommendation ID

Governance Review ID

Audit ID

Replay ID

10.8 Evidence Preservation Requirements

Operational evidence shall preserve:

- Incident records
- Change records
- Release records
- Approval records
- Security investigations
- Governance reviews

10.9 Log Storage Requirements

Production Logs

Centralized Storage

Security Logs

Restricted Storage

Audit Logs

Immutable Storage

10.10 Access Controls

Access shall follow:

- Least Privilege
- Need-to-Know
- Segregation of Duties
- Audit Monitoring

10.11 Retention Requirements

Operational Logs:

12 Months Minimum

Security Logs:

Per Security Architecture

Audit Logs:

Per Compliance Requirements

10.12 Evidence Integrity Controls

- Tamper Detection
- Hash Verification
- Retention Protection
- Backup Protection

10.13 Metrics

- Logging Coverage
- Missing Log Events
- Log Integrity Validation Rate
- Evidence Availability

10.14 Audit Requirements

Auditors must be able to reconstruct:

- User Actions
- System Actions
- Governance Actions
- Security Events

10.15 Traceability

RTM References:

- LOG-001
- LOG-002
- AUD-LOG-001
- REP-001

10.16 Section Summary

This section defines operational logging standards, evidence retention controls, traceability requirements, and audit-ready evidence management practices.

Section 11 – Capacity & Performance Management

11.1 Purpose

Establish capacity planning and performance management processes ensuring sustainable platform operation and scalability.

11.2 Objectives

- Prevent resource exhaustion.
- Support growth.
- Maintain SLA compliance.
- Enable proactive planning.
- Reduce operational risk.

11.3 Scope

Applies to:

- Compute resources
- Storage resources
- Database resources
- Search resources
- API resources
- Network resources

11.4 Capacity Management Principles

- Forecast Before Expansion
- Measure Before Scaling
- Automate Where Possible
- Maintain Operational Resilience

11.5 Capacity Planning Lifecycle

Measure

→ Analyze

→ Forecast

→ Plan

→ Approve

→ Implement

→ Validate

11.6 Capacity Planning Inputs

- Growth forecasts
- Customer adoption trends
- Usage analytics
- Incident history
- Performance metrics
- Release roadmap

11.7 Resource Categories

Category A

Compute

Category B

Storage

Category C

Database

Category D

Search Infrastructure

Category E

Networking

11.8 Capacity Thresholds

Warning Threshold

Critical Threshold

Emergency Threshold

Threshold values shall be maintained within operational configuration repositories.

11.9 Performance Monitoring Areas

- API Response Time
- Search Latency
- Recommendation Latency
- Database Throughput
- Queue Processing Time
- Infrastructure Utilization

11.10 Scaling Requirements

Support:

- Horizontal Scaling
- Vertical Scaling
- Elastic Capacity Expansion
- Disaster Recovery Capacity

11.11 Capacity Review Cadence

Weekly

Operations Review

Monthly

Capacity Planning Review

Quarterly

Strategic Growth Review

11.12 Performance Testing Requirements

Testing shall validate:

- Expected Load
- Peak Load
- Stress Load
- Failure Conditions

11.13 Metrics

- Resource Utilization
- Growth Rate
- Capacity Headroom
- Scaling Events
- SLA Compliance

11.14 Audit Requirements

Maintain:

- Capacity Plans
- Forecast Reports
- Scaling Records
- Review Evidence

11.15 Traceability

RTM References:

- CAP-001
- CAP-002
- PERF-001
- PERF-002

11.16 Ownership

Primary Owner:

Platform Reliability Team

11.17 Section Summary

This section defines capacity planning, performance monitoring, scaling governance, forecasting requirements, and operational review processes supporting TODP growth.

OPS RUNBOOK V1

Section 12 – Backup, Recovery & Data Protection Operations

12.1 Purpose

Define operational controls governing backup management, recovery procedures, and data protection activities.

12.2 Objectives

- Protect critical data assets.
- Enable reliable recovery.
- Support business continuity.
- Meet compliance obligations.
- Preserve operational resilience.

12.3 Scope

Applies to:

- Databases
- Audit repositories
- Governance records
- Configuration repositories
- Operational evidence stores
- Platform metadata

12.4 Data Protection Principles

- Recoverability
- Integrity
- Confidentiality
- Availability

- Traceability

12.5 Backup Categories

Category A

Full Backups

Category B

Incremental Backups

Category C

Snapshot Backups

Category D

Immutable Backups

12.6 Backup Requirements

Production Databases

Scheduled Backups

Audit Repositories

Protected Backups

Configuration Stores

Version-Controlled Backup

Governance Records

Protected Backup Storage

12.7 Recovery Objectives

Recovery Time Objective (RTO)

Defined by Business Impact Classification

Recovery Point Objective (RPO)

Defined by Data Classification and Criticality

12.8 Backup Lifecycle

Backup Creation

→ Verification

→ Encryption

→ Replication

→ Retention

→ Recovery Testing

12.9 Backup Security Controls

- Encryption at Rest
- Encryption in Transit
- Access Control
- Immutable Storage
- Integrity Validation

12.10 Recovery Testing Requirements

Testing Frequency:

Quarterly Recovery Validation

Annual Full Recovery Exercise

12.11 Recovery Validation Requirements

Validate:

- Data Integrity
- Application Functionality
- Audit Availability
- Governance Record Availability

12.12 Retention Requirements

Retention periods shall align with:

- Data Governance V1
- Security Architecture V1
- Compliance Package V1

12.13 Metrics

- Backup Success Rate
- Recovery Success Rate
- Backup Coverage
- Recovery Test Completion Rate

12.14 Audit Requirements

Maintain:

- Backup Logs
- Recovery Test Reports
- Validation Evidence
- Recovery Approvals

12.15 Governance Requirements

Governance review required for:

- Failed recovery tests
- Recovery objective breaches
- Data protection violations

12.16 Traceability

RTM References:

- BAK-001

- BAK-002
- DR-001
- AUD-BAK-001

12.17 Ownership

Primary Owner:

Data Operations Team

Review Authority:

Platform Operations Committee

12.18 Section Summary

This section defines backup operations, recovery controls, recovery validation requirements, data protection measures, governance oversight, and audit obligations supporting TODP operational resilience.

OPS RUNBOOK V1

Section 13 – Disaster Recovery Operations

13.1 Purpose

Establish operational procedures for recovering TODP services following catastrophic failures, regional outages, cyber incidents, or infrastructure disruptions.

13.2 Objectives

- Restore critical services within approved recovery objectives.
- Protect customer and governance data.
- Minimize business disruption.
- Ensure operational resilience.
- Support compliance and audit requirements.

13.3 Scope

Applies to:

- Production platform services
- Infrastructure services
- Data platforms
- Governance repositories
- Audit repositories
- Security systems
- Third-party dependencies

13.4 Disaster Recovery Principles

- Recovery Readiness
- Operational Continuity
- Controlled Execution
- Evidence Preservation
- Governance Oversight

13.5 Disaster Recovery Triggers

- Regional cloud outage
- Data center outage
- Major cyberattack
- Ransomware incident
- Critical infrastructure failure
- Data corruption event
- Extended service unavailability

13.6 Recovery Prioritization

Priority 1

Governance Services

Priority 2

Audit Services

Priority 3

Opportunity Services

Priority 4

Authority Services

Priority 5

Evidence Services

Priority 6

Retrieval Services

Priority 7

Ranking Services

Priority 8

Recommendation Services

Priority 9

Replay Services

13.7 Disaster Recovery Lifecycle

Detection

→ Assessment

→ Disaster Declaration

→ Recovery Activation

→ Service Restoration

→ Validation

→ Business Approval

→ Closure

13.8 Disaster Recovery Roles

Executive Recovery Sponsor

Disaster Recovery Manager

Operations Lead

Security Lead

Infrastructure Lead

Governance Lead

Communications Lead

13.9 Recovery Communication Requirements

Internal Stakeholders

Executive Leadership

Governance Board

Operations Teams

External Stakeholders

Customers

Partners

Regulators (when required)

13.10 Recovery Validation

Validation shall confirm:

- Service availability
- Data integrity
- Security controls
- Audit continuity
- Governance functionality

13.11 Recovery Testing

Quarterly:

Targeted Recovery Exercises

Semiannual:

Cross-Domain Recovery Exercises

Annual:

Full Disaster Recovery Simulation

13.12 Recovery Documentation

Required Records:

- Disaster Declaration
- Recovery Timeline
- Recovery Decisions
- Validation Results
- Executive Approvals

13.13 Metrics

- RTO Achievement
- RPO Achievement
- Recovery Success Rate
- Recovery Exercise Completion Rate

13.14 Audit Requirements

Maintain:

- Recovery Plans
- Exercise Reports
- Validation Records
- Recovery Evidence

13.15 Traceability

RTM References:

- DR-001

- DR-002
- BCP-001
- AUD-DR-001

13.16 Ownership

Primary Owner:

Disaster Recovery Manager

Review Authority:

Platform Operations Committee

13.17 Section Summary

This section defines disaster recovery activation criteria, recovery execution procedures, validation requirements, governance oversight, and audit controls supporting TODP resilience.

OPS RUNBOOK V1

Section 14 – Security Operations Management

14.1 Purpose

Establish operational security procedures protecting TODP assets, services, users, and information throughout production operations.

14.2 Objectives

- Detect security threats.
- Respond to security incidents.
- Protect platform assets.
- Maintain compliance readiness.
- Support enterprise security governance.

14.3 Scope

Applies to:

- Identity systems
- Application security
- Infrastructure security
- Data security
- Network security
- Governance systems
- Audit systems

14.4 Security Operations Principles

- Zero Trust
- Least Privilege
- Continuous Monitoring
- Defense in Depth
- Security by Default

- Evidence Preservation

14.5 Security Operations Functions

Threat Monitoring

Vulnerability Management

Access Management

Security Incident Response

Compliance Monitoring

Security Reporting

14.6 Security Event Categories

Category A

Unauthorized Access

Category B

Privilege Escalation

Category C

Malware Activity

Category D

Data Exposure

Category E

Configuration Weakness

Category F

Policy Violation

14.7 Security Monitoring Requirements

Monitor:

- Authentication events
- Authorization events
- Administrative actions
- Data access events
- API access events
- Infrastructure events

14.8 Security Response Workflow

Detection

→ Validation

→ Classification

→ Containment

→ Investigation

→ Remediation

→ Verification

→ Closure

14.9 Access Governance

Operational requirements:

- Periodic access review
- Role validation
- Privileged access monitoring
- Access certification

14.10 Vulnerability Management

Activities include:

- Vulnerability scanning
- Risk assessment
- Remediation tracking
- Exception management

14.11 Security Reporting

Daily:

Critical Events Review

Weekly:

Security Operations Review

Monthly:

Security Governance Report

Quarterly:

Security Posture Assessment

14.12 Metrics

- Security Event Volume
- Mean Time to Contain
- Vulnerability Remediation Rate
- Access Review Completion Rate

14.13 Audit Requirements

Maintain:

- Security Investigations
- Access Reviews
- Vulnerability Records
- Security Reports

14.14 Compliance Requirements

Support:

- SOC2 Controls
- ISO27001 Controls
- Internal Security Policies

14.15 Traceability

RTM References:

- SEC-001
- SEC-002
- SEC-003
- AUD-SEC-001

14.16 Ownership

Primary Owner:

Security Operations Team

Review Authority:

Chief Information Security Officer

14.17 Section Summary

This section defines security monitoring, incident response, vulnerability management, access governance, reporting requirements, and audit controls supporting TODP security operations.

Section 15 – Operational Communications & Stakeholder Management

15.1 Purpose

Define communication processes supporting operational transparency, incident coordination, stakeholder engagement, and governance reporting.

15.2 Objectives

- Ensure timely communication.
- Reduce information gaps.
- Support coordinated response.
- Maintain stakeholder confidence.
- Provide operational transparency.

15.3 Scope

Applies to:

- Operational incidents
- Security events
- Planned maintenance
- Disaster recovery activities
- Governance reviews

- Service status reporting

15.4 Communication Principles

- Accuracy
- Timeliness
- Consistency
- Accountability
- Transparency

15.5 Stakeholder Groups

Internal

Executive Leadership

Product Teams

Operations Teams

Security Teams

Governance Teams

External

Customers

Partners

Auditors

Regulators

15.6 Communication Categories

Operational Notifications

Incident Communications

Maintenance Notifications

Security Communications

Governance Reports

Executive Briefings

15.7 Incident Communication Lifecycle

Incident Declared

→ Stakeholder Notification

→ Status Updates

→ Resolution Notification

→ Postmortem Communication

15.8 Maintenance Communications

Required Elements:

- Maintenance identifier
- Scheduled window
- Expected impact
- Rollback approach
- Contact information

15.9 Executive Communications

Required for:

- SEV-1 incidents
- Regulatory exposure
- Major outages
- Security breaches
- Disaster recovery activation

15.10 Customer Communications

Communications shall include:

- Impact summary

- Service status
- Expected resolution timeline
- Follow-up actions

15.11 Communication Channels

- Email
- Incident Management Platform
- Collaboration Platform
- Status Portal
- Executive Briefings

15.12 Reporting Cadence

Daily:

Operations Summary

Weekly:

Operational Health Report

Monthly:

Governance Operations Report

Quarterly:

Executive Operations Review

15.13 Metrics

- Communication Timeliness
- Stakeholder Satisfaction
- Reporting Compliance
- Notification Accuracy

15.14 Audit Requirements

Maintain:

- Notifications
- Communications Logs
- Stakeholder Reports
- Approval Records

15.15 Traceability

RTM References:

- COM-001
- COM-002
- GOV-COM-001

15.16 Ownership

Primary Owner:

Operations Communications Manager

Review Authority:

Platform Operations Committee

15.17 Section Summary

This section defines operational communication requirements, stakeholder engagement standards, reporting obligations, and governance communication controls.

OPS RUNBOOK V1

Section 16 – Operational Compliance & Audit Readiness

16.1 Purpose

Establish operational controls ensuring continuous compliance readiness and audit preparedness across TODP operations.

16.2 Objectives

- Demonstrate operational compliance.
- Maintain audit evidence.
- Support regulatory assessments.
- Enable governance transparency.
- Reduce compliance risk.

16.3 Scope

Applies to:

- Operations processes
- Security operations
- Change management
- Incident management
- Data operations
- Governance activities
- Audit activities

16.4 Compliance Principles

- Continuous Readiness
- Evidence-Based Compliance
- Traceability
- Accountability
- Repeatability

16.5 Compliance Framework Alignment

Supported Frameworks:

- SOC2
- ISO27001
- Internal Governance Standards

- Enterprise Procurement Reviews

16.6 Audit Readiness Components

Control Documentation

Operational Records

Evidence Repositories

Governance Records

Training Records

Review Records

16.7 Operational Compliance Reviews

Monthly:

Control Validation Review

Quarterly:

Compliance Assessment

Annual:

Comprehensive Readiness Review

16.8 Evidence Requirements

Required Evidence:

- Incident Records
- Change Records
- Access Reviews
- Security Reviews
- Backup Validation
- Recovery Exercises
- Governance Decisions

16.9 Audit Support Procedures

Audit Request

→ Evidence Collection

→ Validation

→ Submission

→ Auditor Review

→ Closure

16.10 Compliance Exceptions

All exceptions require:

- Risk Assessment
- Business Justification

- Governance Approval
- Expiration Date
- Remediation Plan

16.11 Operational Control Testing

Control testing shall validate:

- Design Effectiveness
- Operating Effectiveness
- Evidence Availability

16.12 Metrics

- Control Compliance Rate
- Audit Finding Count
- Evidence Completeness
- Review Completion Rate

16.13 Governance Requirements

The Platform Operations Committee shall review:

- Audit findings
- Compliance risks
- Control failures
- Exception requests

16.14 Audit Requirements

Maintain:

- Audit Packages
- Evidence Inventories
- Review Records
- Remediation Plans

16.15 Traceability

RTM References:

- CMP-001
- CMP-002
- AUD-001
- AUD-002

16.16 Ownership

Primary Owner:

Compliance & Governance Team

Review Authority:

Product Governance Board

16.17 Section Summary

This section establishes operational compliance governance, audit readiness controls, evidence management requirements, review procedures, and continuous compliance obligations supporting enterprise-grade TODP operations.

OPS RUNBOOK V1

Section 17 – Vendor & Third-Party Operations Management

17.1 Purpose

Establish operational governance and management procedures for third-party vendors, cloud providers, software providers, data providers, and operational service partners supporting TODP.

17.2 Objectives

- Maintain operational continuity across external dependencies.
- Reduce third-party operational risk.
- Ensure contractual service compliance.
- Support security and compliance requirements.
- Establish accountability for vendor-managed services.

17.3 Scope

Applies to:

- Cloud infrastructure providers
- Identity providers

- Monitoring providers
- Security tooling providers
- Data service providers
- Communication service providers
- Managed service providers

17.4 Third-Party Governance Principles

- Risk-Based Oversight
- Continuous Monitoring
- Service Accountability
- Contractual Compliance
- Operational Transparency

17.5 Vendor Classification

Tier 1

Mission Critical Vendors

Tier 2

Business Critical Vendors

Tier 3

Operational Vendors

Tier 4

Low-Risk Vendors

17.6 Vendor Lifecycle

Vendor Assessment

- Risk Review
- Approval
- Onboarding
- Monitoring
- Periodic Review
- Renewal or Termination

17.7 Operational Requirements

All critical vendors shall provide:

- Service commitments
- Security commitments
- Support procedures
- Escalation contacts
- Incident reporting processes

17.8 Vendor Monitoring

Monitor:

- Availability
- SLA Compliance

- Security Events
- Service Performance
- Contract Obligations

17.9 Vendor Incident Management

Third-Party Incident

- Validation
- Impact Assessment
- Escalation
- Resolution Tracking
- Closure

17.10 Business Continuity Requirements

Critical vendors must demonstrate:

- Backup capabilities
- Recovery capabilities
- Continuity planning
- Incident response readiness

17.11 Vendor Review Cadence

Monthly:

Operational Review

Quarterly:

Risk Review

Annually:

Vendor Assessment

17.12 Metrics

- Vendor SLA Compliance
- Vendor Incident Volume
- Vendor Risk Rating
- Vendor Availability

17.13 Audit Requirements

Maintain:

- Vendor Assessments
- Contracts
- SLA Reports
- Risk Reviews
- Incident Records

17.14 Compliance Requirements

Vendor operations must align with:

- Security Architecture V1
- Compliance Package V1
- Data Governance V1

17.15 Traceability

RTM References:

- VND-001
- VND-002
- OPS-VND-001
- AUD-VND-001

17.16 Ownership

Primary Owner:

Vendor Management Function

Review Authority:

Platform Operations Committee

17.17 Section Summary

This section defines operational governance, monitoring requirements, risk oversight, continuity requirements, and audit controls for third-party service providers supporting TODP.

Section 18 – Operational Metrics, Reporting & Service Review Management

18.1 Purpose

Establish measurement, reporting, and review mechanisms used to evaluate operational effectiveness and service performance.

18.2 Objectives

- Support data-driven decision making.
- Measure operational performance.
- Identify operational risks.
- Enable continuous improvement.
- Demonstrate service reliability.

18.3 Scope

Applies to:

- Service operations
- Security operations
- Governance operations
- Platform reliability operations
- Incident management
- Change management

18.4 Reporting Principles

- Accuracy
- Consistency
- Timeliness
- Traceability
- Actionability

18.5 Reporting Hierarchy

Operational Reports

Management Reports

Governance Reports

Executive Reports

Board Reports

18.6 Operational KPI Categories

Availability KPIs

Performance KPIs

Security KPIs

Governance KPIs

Compliance KPIs

Customer Experience KPIs

18.7 Core Operational Metrics

Availability

Service Uptime

Incident Metrics

MTTR

MTTD

Incident Volume

Change Metrics

Change Success Rate

Rollback Rate

Security Metrics

Security Event Volume

Remediation Performance

Governance Metrics

Review Completion Rate

Audit Findings

18.8 Reporting Cadence

Daily

Operational Dashboard

Weekly

Operations Summary

Monthly

Service Review Package

Quarterly

Executive Review Package

18.9 Service Review Meetings

Review Topics:

- Availability
- Performance
- Incidents
- Capacity
- Security
- Risks
- Compliance

18.10 Continuous Improvement Reviews

Inputs:

- Incident Trends
- Problem Records
- Audit Findings
- Customer Feedback
- Security Reviews

18.11 Executive Reporting Requirements

Reports shall include:

- KPI Performance
- Major Risks
- SLA Compliance
- Strategic Concerns

18.12 Metrics Quality Requirements

Metrics shall be:

- Verifiable
- Repeatable
- Traceable
- Auditable

18.13 Audit Requirements

Maintain:

- KPI Definitions
- Reports
- Review Minutes
- Action Registers

18.14 Traceability

RTM References:

- KPI-001
- KPI-002
- OPS-MET-001
- AUD-KPI-001

18.15 Ownership

Primary Owner:

Platform Operations Committee

18.16 Section Summary

This section establishes operational measurement standards, reporting structures, service review processes, governance reporting requirements, and continuous improvement mechanisms.

OPS RUNBOOK V1

Section 19 – Operational Knowledge Management & Runbook Governance

19.1 Purpose

Establish governance and maintenance requirements for operational knowledge assets, procedures, playbooks, and runbooks supporting TODP operations.

19.2 Objectives

- Preserve operational knowledge.
- Improve operational consistency.
- Reduce dependency on individual personnel.
- Accelerate incident response.
- Support audit readiness.

19.3 Scope

Applies to:

- Runbooks
- Operational procedures
- Recovery procedures
- Security playbooks
- Escalation guides
- Architecture references

19.4 Knowledge Management Principles

- Documentation First
- Operational Consistency
- Version Control
- Continuous Improvement
- Controlled Access

19.5 Knowledge Asset Categories

Category A

Operations Runbooks

Category B

Security Playbooks

Category C

Recovery Procedures

Category D

Governance Procedures

Category E

Architecture References

19.6 Runbook Requirements

Each runbook shall contain:

- Purpose
- Scope
- Preconditions
- Procedure Steps
- Escalation Guidance
- Validation Steps

- References

19.7 Documentation Lifecycle

Create

→ Review

→ Approve

→ Publish

→ Maintain

→ Retire

19.8 Review Requirements

Quarterly:

Operational Review

After Major Incident:

Immediate Review

After Major Release:

Validation Review

19.9 Version Control Requirements

All knowledge assets shall maintain:

- Version Number

- Change History
- Approval Records
- Review Dates

19.10 Knowledge Accessibility

Authorized personnel shall have access to:

- Current procedures
- Escalation guides
- Recovery procedures
- Service ownership records

19.11 Training Requirements

Operational staff shall receive training on:

- Runbook usage
- Incident procedures
- Recovery procedures
- Security response procedures

19.12 Metrics

- Documentation Coverage
- Review Completion Rate
- Runbook Accuracy Rate

- Training Completion Rate

19.13 Audit Requirements

Maintain:

- Runbooks
- Review Records
- Approval Records
- Training Evidence

19.14 Traceability

RTM References:

- DOC-001
- DOC-002
- OPS-DOC-001
- AUD-DOC-001

19.15 Ownership

Primary Owner:

Operations Excellence Function

Review Authority:

Platform Operations Committee

19.16 Section Summary

This section establishes governance, maintenance requirements, review obligations, and audit controls for operational knowledge assets and runbooks.

OPS RUNBOOK V1

Section 20 – Operational Continuous Improvement & Maturity Management

20.1 Purpose

Establish the framework used to evaluate, improve, and mature operational capabilities across TODP services.

20.2 Objectives

- Improve operational effectiveness.
- Increase platform reliability.
- Reduce operational risk.
- Enhance service quality.
- Drive operational excellence.

20.3 Scope

Applies to:

- Operations processes
- Security operations
- Incident management
- Change management
- Governance processes
- Reliability engineering activities

20.4 Continuous Improvement Principles

- Measure Performance
- Identify Gaps
- Prioritize Improvements
- Implement Corrective Actions
- Validate Outcomes

20.5 Improvement Sources

- Incident Reviews
- Problem Reviews
- Audit Findings
- Compliance Assessments
- Security Reviews
- Customer Feedback
- Performance Reports

20.6 Improvement Lifecycle

Observation

→ Assessment

→ Prioritization

→ Planning

→ Implementation

→ Validation

→ Closure

20.7 Operational Maturity Domains

Domain A

Governance

Domain B

Reliability

Domain C

Security

Domain D

Compliance

Domain E

Automation

Domain F

Operational Excellence

20.8 Maturity Assessment Model

Level 1

Initial

Level 2

Managed

Level 3

Defined

Level 4

Measured

Level 5

Optimized

20.9 Improvement Governance

The Platform Operations Committee shall:

- Review improvement opportunities
- Prioritize initiatives
- Track remediation progress
- Approve maturity objectives

20.10 Improvement Planning

Plans shall include:

- Objective
- Business Value
- Risk Reduction
- Resource Requirements
- Success Metrics

20.11 Validation Requirements

Completed improvements shall demonstrate:

- Measurable benefit
- Risk reduction
- Operational effectiveness
- Governance compliance

20.12 Annual Operational Assessment

Assessment shall evaluate:

- Process effectiveness
- Technology effectiveness
- Organizational readiness

- Compliance maturity
- Reliability maturity

20.13 Metrics

- Improvement Completion Rate
- Operational Maturity Score
- Risk Reduction Achievement
- Reliability Improvement Rate

20.14 Audit Requirements

Maintain:

- Improvement Registers
- Assessment Reports
- Validation Evidence
- Governance Decisions

20.15 Traceability

RTM References:

- IMP-001
- IMP-002
- MAT-001
- AUD-IMP-001

20.16 Ownership

Primary Owner:

Head of Platform Operations

Review Authority:

Product Governance Board

Approval Authority:

Executive Leadership Team

20.17 Section Summary

This section establishes the continuous improvement framework, operational maturity model, governance oversight processes, measurement standards, and validation controls supporting long-term operational excellence within TODP.

END OF OPS RUNBOOK V1